

Institute of Actuaries of India

Subject ST5 – Finance and Investment A

May 2014 Examinations

INDICATIVE SOLUTIONS

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1 :**i.**

- Viewing the scheme as a going concern, sufficient funds should be held in the pension fund to ensure that the contribution rate does not have to rise markedly in the future. So the primary purpose of the fund is to stabilise the long-term contribution rate of the employer and the employee.
- If the scheme were to wind up tomorrow, the fund should be sufficient to meet all the liabilities in the future that have already been guaranteed. So the assets held should have a minimum value.

[2]**ii.**

- A pension is likely to be a major component of any individual's wealth.
- Unlike stocks and shares, it is virtually impossible to diversify one's pension arrangements in order to spread the risk of bad luck, fraud or mismanagement.
- Whereas a company can afford to plan for the short-term to medium-term future, a pension scheme has responsibilities to pensioners which might not crystallise until they retire in the distant future. Once that time comes, the fund has to be in a position to meet its commitments in full.

Although a pension fund can invest in a diversified portfolio of securities, it is usually managed by one pension administrator or Investment Management Company. If an employee spends her entire career with one employer, and participates in that employer's contracted out pension arrangement, then her entire pension liability is secured by the employer's pension scheme. Although some larger employers split their pension fund across two or more investment managers, this is still relatively uncommon.

[4]

iii.

Returns on investments		
	Investment income	3,554
	Change in asset value	15,506
	Investment expenses	-1,500
	Net return on Investments	17,560
Fund statement		
Inflows	Contribution received	4,041
	Transfer in	378
	Other income of the scheme	385
	Total Inflow	4,804
Outflows	Benefits Payable	6,954
	Leavers	7,452
	Other Payments from scheme	255
	Administrative expenses	267
	Total Outflow	14,928
	Net Outflow	10,124
Opening fund balance		89,474
	Net increase	7,436
Closing Fund balance		96,910

[3]

[Total Marks-11]

Solution 2 :**i.**

- An ALM study can predict a range of projected outcomes for the surplus of asset less liabilities for each of a number of different asset strategies. However, it is then difficult to determine which strategy is “best”.
- Typically equities are expected to give a higher long-term return than bonds. Thus strategies with high equity proportions will normally seem to be the “best”, in terms of leading to a higher average projected surplus or a lower projected average contribution rate, when the outcomes are discounted back to the current date.
- However, the variance of projected outcomes is also likely to be higher for strategies with high equity proportions.
- It can therefore be difficult to determine whether the benefits of additional return more than offset the additional risk, and hence which investment strategy is optimal for the scheme.

[3]**ii.**

- The use of deflators ensures that the cash flows emerging in a particular projection are weighted according to the economic characteristics emerging in that projection, so reflecting the fact that the values placed on cash flows may be state-dependent in practice.
- Deflators place arbitrage-free values on sets of cash flows
- Deflators are consistent with economic observables – *i.e.* current market prices – and so can place market-consistent values on liabilities.
- They can also be used to assess the impact of different investment strategies on all of the different stakeholders in a pension scheme – *e.g.* employer, the shareholders of the sponsoring company, the pensioners, *etc.*
- Finally, they can take account of the options that are inherent in the management of a pension scheme. In particular, the option held by the company’s shareholders to vary the level of pension benefits and/or contributions.
- In particular, by allowing explicitly for the shareholders option to default on future pension promises, deflators do not take advance credit for the higher expected returns on equities. This is in contrast to a traditional discounted cash flow valuation of liabilities, in which the discount rate is varied to reflect the scheme’s assets.

[5]**[Total Marks-8]**

Solution 3 :**i.**

	Fund at start	250		
	Fund at end	277.59		
	Assumption: return is equally distributed over time period			
	MWRR	6.85%	Per Quarter	
		30.3%	Per annum	
	LIRR			
	In the absence of additional information LIRR is equal to MWRR			
	Return on index			
	1.078125	65%		
	1.0625	26%		
	1.09375	9%		
	7.55%	Per Quarter	0.70%	
	33.8%	Per annum		

[6]

ii.

			Original	Extra	
		1.078125	150	6	
		1.0625	50	2	
		1.09375	50	2	
			269.53	10.38	
		Total	279.91		
	Index return on fund	279.24			
	Trustee's over performance			0.67	
	Fund Manager vs. trustee benchmark				
			Original	Extra	
		1.078125	193.75	10	
		1.0625	50	0	
		1.09375	6.25	0	
			268.8	10.4	
			Total	279.2	
	Fund manager underperformance			-0.68	
	Fund manager sector selection skill				
	Actual fund	277.59			
	Fund manager underperformance	-1.65			

[8]

iii. Comment on the results

- The fund manager under-performed both the index and the trustees' benchmark, but only by a small amount, over a short period of time.
- The under-performance of the actual fund compared with the index, as measured by the linked internal rate of return, was about 0.70%. Measured by the All-Share notional fund (279.238) the under-performance was 1.65.
- The under-performance of the actual fund compared with the trustees' benchmark, as measured by the notional funds, was 2.31 (*i.e.* 279.91 – 277.6).

This may reflect:

- lack of skill
- a more cautious approach to stock selection (*e.g.* lower beta shares)
- bad luck
- The management and transactions costs incurred implementing the chosen investment strategy.
- The sector decision to avoid smaller companies, which performed relatively well, explains most of the under-performance.
- Most of the loss is due to stock selection.

In fact, the stock selection can be broken down into:

- stock selection loss in the IT sector (7.10% actually achieved against 7.81% index return)
- Stock selection loss in the Financial Sector (5.9% actually achieved against 6.3% index return).
- The manager, as he thought, may have particular skill in the selection of IT companies. However, this skill is not confirmed. But really we have too little information to be certain.

Further information required

- What was the final fund value?
- Crucially, we need data over a much longer time period (*e.g.* 5 years) to draw any reliable conclusions.
- What sort of stocks did the manager invest in and what was the rationale for doing so?
- What expenses were incurred by the fund in pursuing the chosen investment policy and how would they compare to those incurred by a completely passive index tracking approach?
- Would it be feasible for the fund to be split so that this manager actively manages investment in larger companies, and either another manager or an index tracking approach is used for smaller companies?

[9]

iv. Individuals are typically **overconfident** about their own skills and insights. Overconfidence arises from:

- Hindsight bias – events that happen will be thought of as having been predictable prior to the event, events that do not happen will be thought of as having been unlikely prior to the event.
- Confirmation bias – people will tend to look for evidence that confirms their point of view (and will tend to dismiss evidence that does not justify it).

[2]

[Total Marks-25]

Solution 4 :

i.

The value of the index after effecting these changes should remain at 6500 as index needs should be affected by the change in price of constituents alone and not by corporate actions. This is done by making appropriate changes in the divisor. While dividend leads to fall in value of index considering the quantum of special dividend it will be adjusted so that the index is not affected or else it will lead to a sharp fall in index value.

[3]

ii.

Company	Share Capital	Face Value	Promoter Stake	CMP	No. of shares	MCAP	Free Float	FF MCAP	Wt %	New no. of shares	Face Value2	New Share Capital	Promoter Stake2	Adj. MCAP	Adj. Price - Current	New Free Float	New Free Float MCAP	Revised Weight%	Price Increase %	MCAP after price change
A	39.94	2.00	39.92%	2,016	19.97	40,260	60.08%	24,188	11.80%	18.97	2.00	37.94	42.02%	37,763	1,991	57.98%	21,895	11.30%	2.0%	22,333
B	4,277.76	5.00	69.23%	323	855.55	2,76,343	30.77%	85,031	41.48%	855.55	5.00	4,277.76	69.23%	2,33,566	273	30.77%	71,868	37.08%	2.0%	73,306
C	601.14	10.00	68.43%	488	60.11	29,336	31.57%	9,261	4.52%	60.11	10.00	601.14	58.43%	29,336	488	41.57%	12,195	6.29%	3.0%	12,561
D	684.03	10.00	58.60%	1,694	68.40	1,15,875	41.40%	47,972	23.40%	75.24	10.00	752.43	58.60%	1,22,715	1,631	41.40%	50,804	26.21%	-5.0%	48,264
E	139.37	2.00	61.75%	1,445	69.69	1,00,695	38.25%	38,516	18.79%	75.69	2.00	151.37	64.78%	1,05,195	1,390	35.22%	37,047	19.12%	5.0%	38,900
Total								2,04,968	100.00%								1,93,809	100.00%		1,95,362

New Index level is $6500 \times 195362 / 193809 = 6552.1$

While special dividends are usually adjusted to keep the value of the index same as quantum is usually large the candidate may not be penalised if he/she has assumed that index value has gone down after the dividend and performed the calculations. In this case the final index value will be 6131.

[17]

[Total Marks-20]

Solution 5 :

- i. Value at Risk is a measure of the maximum expected loss that might be suffered on a portfolio or a holding in a specified timescale and with a specified probability.

[2]

ii.

- VaR calculated using the assumption that stock log returns are distributed normally is found to be an underestimate in practice. This is because short period stock returns show a more fat-tailed distribution.
- The assumption of normality also implies that stock returns are symmetrically distributed. In practice, return distributions may be skewed.
- The calculation assumes that the past experience will be maintained, whereas the future probability distribution may well be different (particularly regarding volatility).
- The VaR figure, once calculated, does not say anything about how bad losses could actually be (and definitely does not specify the worst possible loss).
- VaR ignores any problems relating to market liquidity.

[5]

iii. The following is the portfolio

Name	Holding	Price	Amt.	Wt %
A	1200	234	2,80,800	53.9%
B	1350	178	2,40,300	46.1%
			5,21,100	100.0%

$$\sigma_{AB} = \rho_{AB} \sigma_A \sigma_B$$

$$\text{Portfolio Variance} = \begin{bmatrix} 0.539 & 0.461 \end{bmatrix} \begin{bmatrix} 0.0400 & 0.0288 \\ 0.0288 & 0.0324 \end{bmatrix} \begin{bmatrix} 0.539 \\ 0.461 \end{bmatrix}$$

Standard deviation of portfolio = 18.1%

$$\text{VaR} = 1.65 \times 18.1\% \times 521,100 / (250^{0.5}) = 9,851$$

VaR as a % of portfolio = 1.89%

[4]

iv. The new portfolio is as follows

Name	Holding	Price	Amt.	Wt %
A	1200	234	2,80,800	36.7%
B	1350	178	2,40,300	31.4%
C	1800	136	2,44,800	32.0%
			7,65,900	100.0%

$$\text{Portfolio Variance} = \begin{bmatrix} 0.367 & 0.314 & 0.320 \end{bmatrix} \begin{bmatrix} 0.0400 & 0.0288 & 0.0196 \\ 0.0288 & 0.0324 & 0.0151 \\ 0.0196 & 0.0151 & 0.0196 \end{bmatrix} \begin{bmatrix} 0.367 \\ 0.314 \\ 0.320 \end{bmatrix}$$

Standard deviation of portfolio = 15.8%

VaR = 1.65 x 15.8% x 765,900 / (250^{0.5}) = 12,592

VaR as a % of portfolio = 1.64%

Comment:

Diversification benefit helps to bring down VaR (as a % of portfolio)

[6]

v. The VaR increases to 12592+244800x0.12/365 = 12672

VaR as a % of portfolio = 2.4%

Comment:

Leverage leads to significant increase in VaR (as a % of portfolio)

Full marks to be awarded if trading days taken as 365 instead of 250

[3]

[Total Marks-20]

Solution 6 :

- Property or other non-quoted assets may be difficult to value and not be captured in the book value which states it at historical cost
- A company which has made an acquisition may have significant amount of goodwill which can affect the book value. Adjustments for this may have to be made when doing a valuation exercise.
- Intangibles like brand value, patents, trademarks, distribution network, customer loyalty, switching costs associated with the product, skill of the human capital etc. will not be captured by the book value.
- Different industries have different characteristics – a manufacturing intensive industry like textiles is drastically different from a service industry like information technology or a consumer product company which thrives on brands. Comparing price to book across different sectors make little sense.
- Doesn't give any indication about the competitive intensity prevailing. A company which is in a dwindling industry but a track record of good historical profits may look good on a P/BV basis when reality is the opposite.
- Can get affected by accounting policies. For example a company which follows WDV depreciation may have a significantly different P/B compared to a company which follows SLM depreciation even if both belong to same industry.

[6 Marks]**Solution 7 :**

The CFO's analysis is based on apportionment of the expenses and investment income on the basis of earned premium. His analysis could be as follows:

	Health	Liability	Fire	Total
Earned Premium	1000	500	500	2000
Incurred Claim	900	500	450	1850
Commission	50	25	25	100
Technical Reserves	100	850	150	1100
Other Expenses	75	37.5	37.5	150
U/W Results	-25	-62.5	-12.5	-100
Inv. Income	55	27.5	27.5	110
Insurance Profit	30	-35	15	10

While apportionment of expenses on the basis of earned premium is reasonable applying the same logic to investment income is erroneous as investment income is more a function of technical reserves than earned premium. Technical reserves include claims outstanding and unexpired risk reserves. Liability is long tailed with reporting delay itself running into a few years. So the technical reserves earn investment income till the claim is paid and this can be pretty significant. If we reallocate the investment income on the basis of technical reserves the picture which emerges is as follows:

	Health	Liability	Fire	Total
Earned Premium	1000	500	500	2000
Incurred Claim	900	500	450	1850
Commission	50	25	25	100
Technical Reserves	100	850	150	1100
Other Expenses	75	37.5	37.5	150
U/W Results	-25	-62.5	-12.5	-100
Inv. Income	10	85	15	110
Insurance Profit	-15	22.5	2.5	10

In terms of insurance profit Liability is what contributes most to the profitability. When to pay is as important as how much to pay especially in long tailed businesses like Liability.

Health:

Since the technical reserves are only 10% of earned premium it indicates that the business is short tailed and claims get reported and paid very quickly. So money market instruments, short term fixed deposits of banks etc. can be considered as suitable asset classes.

Fire:

The size of technical reserves with respect to premium indicates that the suitable asset classes could be short term government or corporate bonds, fixed deposits in banks etc.

Liability:

Since claims take quite a while to get reported and to get settled inflation protection is a key factor to consider. Some portion may be allocated to Long term government bonds; Infrastructure bonds etc. while to protect the company from claim inflation a portion needs to be allocated to asset classes like equity, equity mutual funds, index funds etc.

[10 Marks]
